Solution Tutorial I – Lift

Task 1

Determine the aspect ratio AR and taper ratio λ of the configuration with the given values in the data set. Compare the results with those given by the manufacturer and explain the differences.

- Aspect ratio:
  - Calculated: $AR = \frac{b^2}{S_{ref}} = \frac{34.1^2}{122.4} = 9.500$
  - Manufacturer: $AR = 9.39$

  Difference of the values is due to the difference in reference areas used. For some reason, the manufacturer uses the Boeing reference area in order to calculate the AR. As the Boeing area is slightly greater, the resulting AR is smaller than the one we calculated here.
- Taper ratio:
  - Calculated: $\lambda = \frac{c_{tip}}{c_{root}} = \frac{1.5}{6.07} = 0.2471$
  - Manufacturer: $\lambda = 0.247$

Task 2

Estimate the lift coefficient slope $C_{L\alpha}$ of the finite wings according to the lifting line theory.

- Lift coefficient slope (lifting line theory):

$$C_{L\alpha} = \frac{2\pi \cdot AR}{2 + AR} = \frac{2\pi \cdot 9.39}{2 + 9.39} = 5.180\ /rad$$

Task 3

For a more accurate result, $C_{L\alpha}$ is to be calculated using the equation of Polhamus at the cruise Mach number of $M = 0.78$. Name a possible cause for the difference to the previously calculated value.

- Lift coefficient slope (Polhamus method):

$$C_{L\alpha} = \frac{2\pi \cdot \mathrm{AR}}{2 + \sqrt{\frac{\mathrm{AR}^2 \cdot \beta^2}{\mathrm{K}^2} \cdot \left(1 + \frac{\tan^2\Lambda_{50\%}}{\beta^2}\right) + 4}}$$

With:

- $\beta = \sqrt{1 - M^2} = \sqrt{1 - 0.78^2} = 0.6258$
- $K = 1 + 0.75 \cdot \left(\frac{t}{c}\right) = 1 + 0.75 \cdot (11.75\%) = 1.088$ (for rel. thickness at kink)

  Here, also the mean relative thickness of the wing could be used.
- $\tan(\Lambda_{50\%}) = \tan(\Lambda_{25\%}) - \frac{4}{AR} \cdot \left[\frac{50-25}{100} \cdot \frac{1-\lambda}{1+\lambda}\right] =$

$$= \tan(0.4356\, rad) - \frac{4}{9.39} \cdot \left[\frac{50-25}{100} \cdot \frac{1-0.2471}{1+0.2471}\right] = 0.4011$$

Therefore:

$$C_{L\alpha} = \frac{2\pi \cdot 9.39}{2 + \sqrt{\frac{9.39^2 \cdot 0.6258^2}{1.088^2} \cdot \left(1 + \frac{0.4011^2}{0.6258^2}\right) + 4}} = 6.766\ /rad$$

The value is different to the one calculated with lifting line theory in task 2 due to compressibility effects we encounter at a cruise Mach number of 0.78. The equation for the lifting line theory is only applicable for $M \ll M_{crit}$.

Task 4

Plot $C_{L\alpha}$ against the Mach number for the region $M = 0.2 \ldots 0.8$ with the help of one calculation method (choose the most appropriate one). Extend the plot for the transonic regime ($M = 0.8 \ldots 0.95$) and explain the curve path.

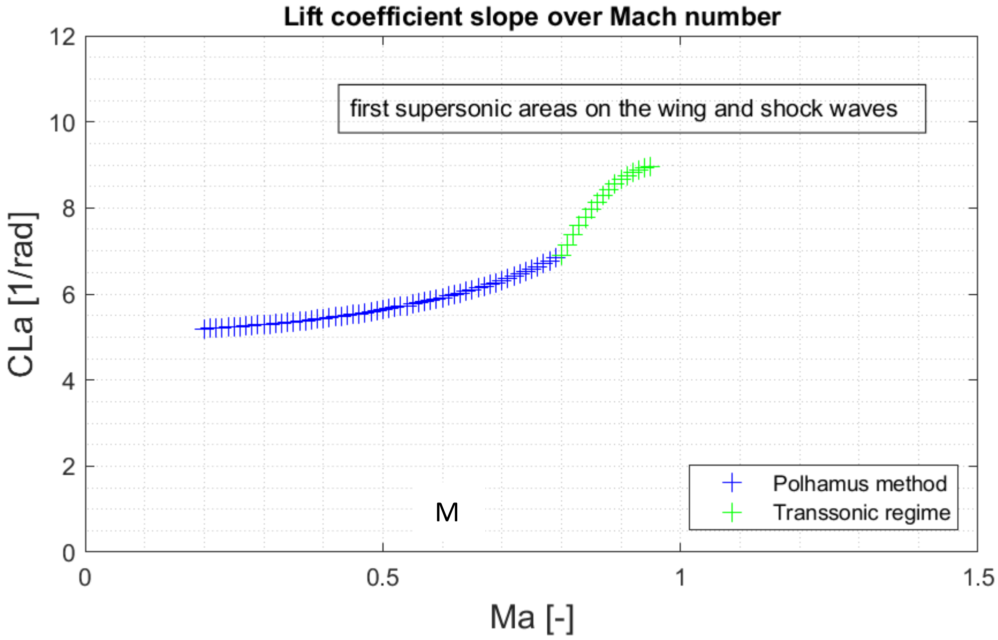
M_{cr} is given to be 0.8 (see problem formulation). Hence, for $M = 0.2 \ldots 0.8$ Polhamus can be applied:

$$C_{L\alpha} = \frac{2\pi \cdot AR}{2 + \sqrt{\frac{AR^2 \cdot \beta^2}{K^2} \cdot \left(1 + \frac{\tan^2 \Lambda_{50\%}}{\beta^2}\right) + 4}}$$

with $\beta = \sqrt{1 - (0.2)^2} \ldots \sqrt{1 - (0.8)^2}$

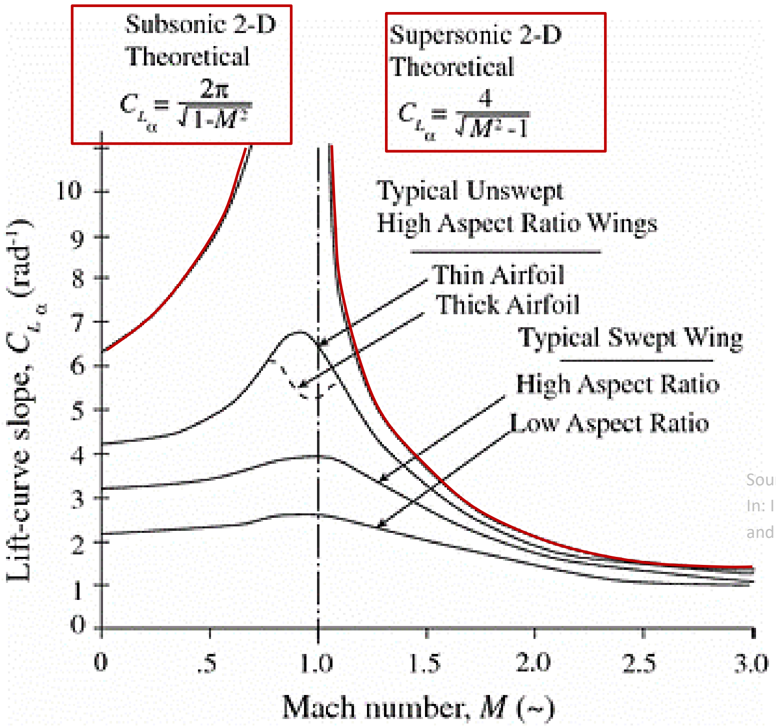
$M\ [-]$	$C_{L\alpha}\ [1/rad]$
0.2	5.197
0.3	5.290
0.4	5.431
0.5	5.632
0.6	5.913
0.7	6.313
0.8	6.907

Plot:



Explanation for the transonic regime curve: we encounter first supersonic areas on the wing and shock waves in that regime (compressibility effects).

Also compare:



Source: Vos R., Farokhi S. (2015) Transonic Similarity Laws. In: Introduction to Transonic Aerodynamics. Fluid Mechanics and Its Applications, vol 110. Springer, Dordrecht

Task 5

Plot the lift curve (C_L vs. α) for low Mach number (use the estimation of $C_{L\alpha}$ from Task 2 and given geometrical data in the data set). Assume an aircraft stall angle of $\alpha_{max} = 10°$.

In order to plot the lift curve, three parameters are necessary:

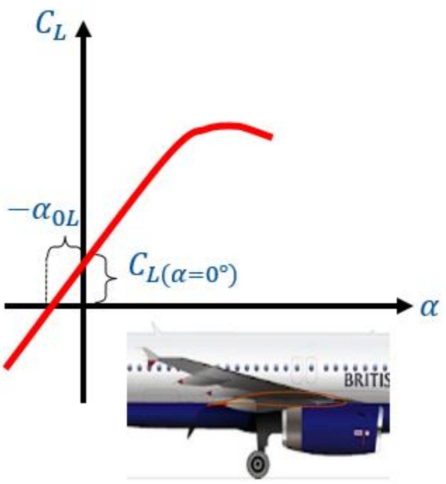
a) The **curve gradient** in 1/rad or 1/deg
b) The **curve shift** from the y-axis due to airfoil camber and/or wing incidence
c) The **curve form** at max. angle of attack

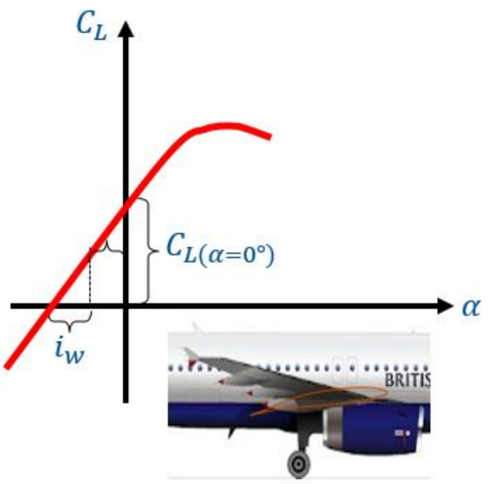
a) **Curve gradient** in 1/rad or 1/deg of Task 2 (lifting line):

$$C_{L\alpha} = \frac{2\pi \cdot AR}{2 + AR} = \cdots = 5.180\ /rad$$

b) **Curve shift** from the y-axis due to airfoil camber and wing incidence:

- *Airfoil camber*: airfoil angle of attack at $C_L = 0 \Longrightarrow \alpha_{0L} = -3°$ (in aircraft dataset: -0.0524 rad)
- *Wing incidence* with respect to fuselage: $i_W = 3.66°$ (in aircraft dataset: $+0.0639$ rad)





The two effects are superimposed.

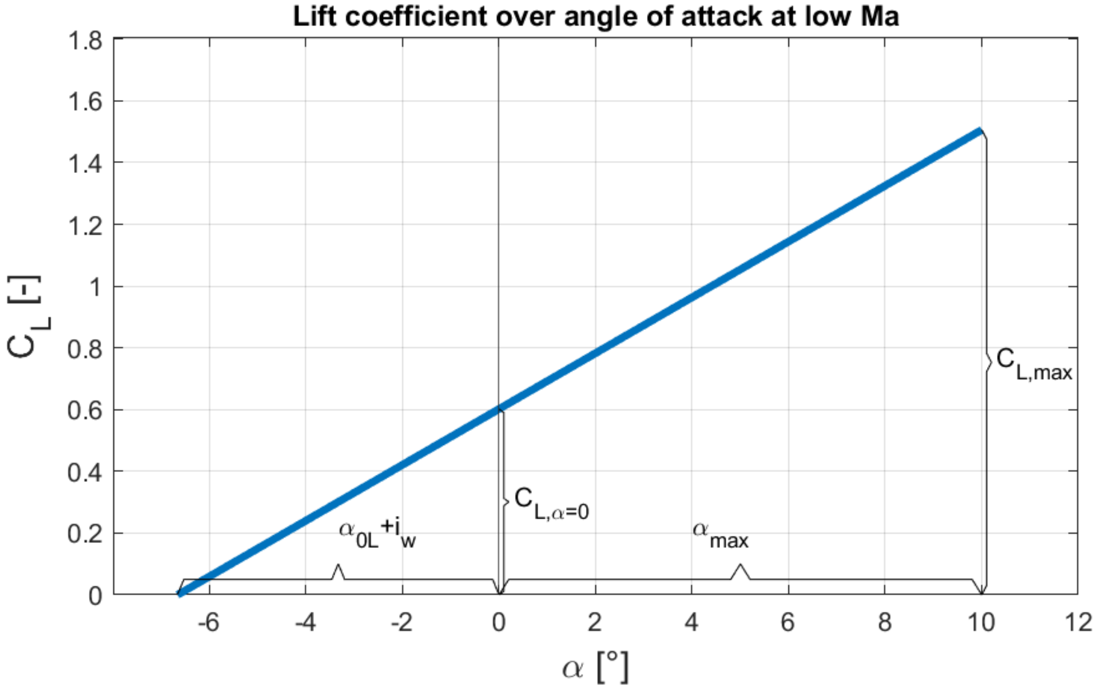
$\alpha = 0° \Longrightarrow \quad C_{L(\alpha=0°)} = C_{L\alpha} \cdot (|-3°| + 3.66°) = 5.180\ /\mathrm{rad} \cdot 0.1163\ \mathrm{rad} = 0.6024$

c) **Curve form** at max. angle of attack:

- Here a linear overall form is assumed. The exact form at stall and in the post-stall region would require further precise airfoil and wing data.
- α_{max} is given to be $10°$ $(= 0.1745\ rad)$.
- At this angle of attack: $C_{L,max} = C_{L(\alpha=0°)} + C_{L\alpha}\ \alpha_{max}$

$$= 0.6024 + 5.180\ /\mathrm{rad} \cdot 0.1745\ \mathrm{rad} \approx 1.506$$

- Draw curve:



Task 6

Determine the reference 1g stalling speed V_{S1g} at maximum take-off weight in clean configuration during level flight at sea level conditions. Draw the free body diagram of the aircraft in this situation.

- Free-body diagram:

- From the diagram:

  Condition for level flight:

$$L = w$$

$$\Longrightarrow \frac{1}{2}\rho V^2 \cdot C_L \cdot S_{ref} = mg$$

  In described flight condition:

$$\frac{1}{2}\rho {V_S}^2 \cdot C_{L,max} \cdot S_{ref} = m_{TO} \cdot g$$

$$\Longrightarrow V_S = \sqrt{\frac{2 \cdot m_{TO} \cdot g}{\rho \cdot C_{L,max} \cdot S_{ref}}}$$

$$V_S = \sqrt{\frac{2 \cdot 73500 \cdot 9.81}{1.225 \cdot 1.506 \cdot 122.4}} = 79.91 \frac{m}{s}$$

Task 7

Determine the aircraft angle of attack α_C with a cruise speed of $M_C = 0.78$ at 90% of the maximum take-off weight at an altitude of $h = 12km$. Assume $\alpha_{max} = 6°$.

In order to determine α_C, the **lift curve is required**.

Therefore, repeat Task 5, where following parameters are necessary:

a) The curve gradient in 1/rad or 1/deg: Polhamus required as $M_C < M_{cr}$ (see Task 3)

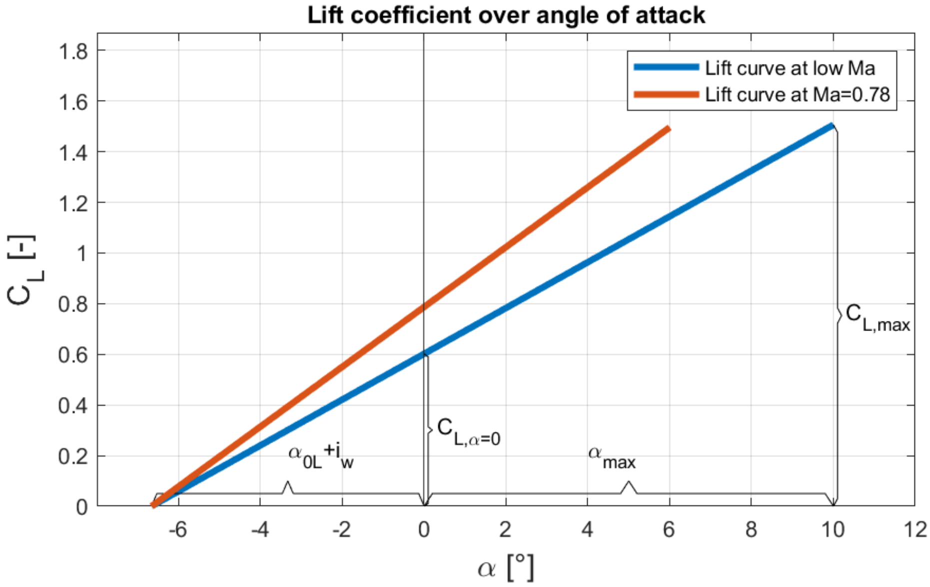
$$C_{L\alpha} = \ldots = 6.766\ /rad$$

b) The curve shift from the y-axis due to airfoil camber and/or wing incidence

$$\alpha = 0° \Longrightarrow \quad \mathrm{C}_{\mathrm{L}(\alpha=0°)} = C_{L\alpha} \cdot (|-3°| + 3.66°) = 6.766\ /\mathrm{rad} \cdot 0.1163\ \mathrm{rad} = 0.7869$$

c) The curve form at max. angle of attack of 6° = 0.1047 rad

$$C_{L,max} = C_{L(\alpha=0°)} + C_{L\alpha}\ \alpha_{max} = 0.7869 + 6.766\ /\mathrm{rad} \cdot 0.1047\ \mathrm{rad} \approx 1.495$$



The next step is to calculate the lift coefficient for this setting.

- From standard atmosphere: $a = 295$ m/s & $\rho = 0.311\ kg/m^3$ @ cruise altitude

$$\Longrightarrow V = M \cdot a = 0.78 \cdot 295 = 230.1\ \mathrm{m/s}$$

- Condition for level flight: $L = w$

$$\Longrightarrow \frac{1}{2}\rho \cdot V^2 \cdot C_{L,C} \cdot S_{ref} = 0.90 \cdot m_{TO} \cdot g$$

$$\Longrightarrow C_{L,C} = \frac{0.90 \cdot m_{TO} \cdot g}{\frac{1}{2}\rho \cdot V^2 \cdot S_{ref}}$$

$$\Longrightarrow C_{L,C} = \frac{0.90 \cdot 73500 \cdot 9.81}{\frac{1}{2} 0.311 \cdot 230.1^2 \cdot 122.4} = 0.6440$$

- Read from diagram (or calculate):$\Longrightarrow \alpha_C \approx -1.21°$